

chapter 10

What is Util Package

The `java.util` package is one of Java's core utility packages. It contains utility classes, data structure frameworks, event models, date and time facilities, and internationalization support. Key components include the Collection Framework classes (like `ArrayList`, `HashMap`), legacy utility classes (like `Vector`, `Hashtable`), and helper classes like `Scanner`, `Random`, and `Date`.

What is Collection Framework

The Java Collection Framework (JCF) is a unified architecture for representing, storing, and manipulating groups of objects. It provides pre-packaged data structures and algorithms, reducing programming effort while increasing performance and code readability. It consists of:

- **Interfaces:** Define abstract data types (e.g., `List`, `Set`, `Map`).
- **Implementations:** Concrete classes that implement these interfaces (e.g., `ArrayList`, `HashSet`, `HashMap`).
- **Algorithms:** Utility methods for searching, sorting, and transforming collections (found in the `Collections` class).

Collection vs Collections

- **Collection (Interface):** The root interface in the Java Collection Framework hierarchy (`java.util.Collection`). It defines fundamental operations for groups of objects (e.g., `add()`, `remove()`, `size()`).
- **Collections (Utility Class):** A helper class (`java.util.Collections`) consisting exclusively of static methods that operate on or return collections (e.g., sorting, reversing, searching).

Java

```
import java.util.*;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        // Collection interface reference holding an ArrayList implementation
```

```
        Collection<String> names = new ArrayList<>();
```

```

names.add("Zara");

names.add("Alex");


// Collections utility class sorting the list

Collections.sort((List<String>) names);


System.out.println(names); // Output: [Alex, Zara]

}

}

```

Collection Framework

The **Java Collection Framework (JCF)** is a unified architecture representing and manipulating collections. It consists of:

- **Interfaces:** Collection, List, Set, Queue, Map.
- **Implementations (Classes):** ArrayList, LinkedList, HashSet, HashMap, etc.
- **Algorithms:** Static methods provided by Collections or Arrays.

Java

```

import java.util.*;


public class Main {

    public static void main(String[] args) {

        // Using List interface from the Collection Framework

        List<Integer> numbers = new ArrayList<>();

        numbers.add(10);

        numbers.add(20);


        for (int num : numbers) {

```

```
        System.out.println(num);
    }
}
}
```

Map Interface

The **Map interface** (java.util.Map) stores data in **key-value pairs**. Key characteristics:

- Keys must be unique.
- Each key maps to exactly one value.
- It does **not** inherit from the Collection interface, though it is part of the Collection Framework.

Java

```
import java.util.*;
```

```
public class Main {

    public static void main(String[] args) {

        Map<Integer, String> studentMap = new HashMap<>();

        studentMap.put(101, "Alice");

        studentMap.put(102, "Bob");


        // Fetching value using key

        System.out.println("ID 101: " + studentMap.get(101)); // Output: Alice

    }

}
```